AI OPEN
DATA CENTER:
Power Distribution



WHAT THE AUTOMOTIVE INDUSTRY CAN TEACH US ABOUT 800 VDC DATA CENTERS

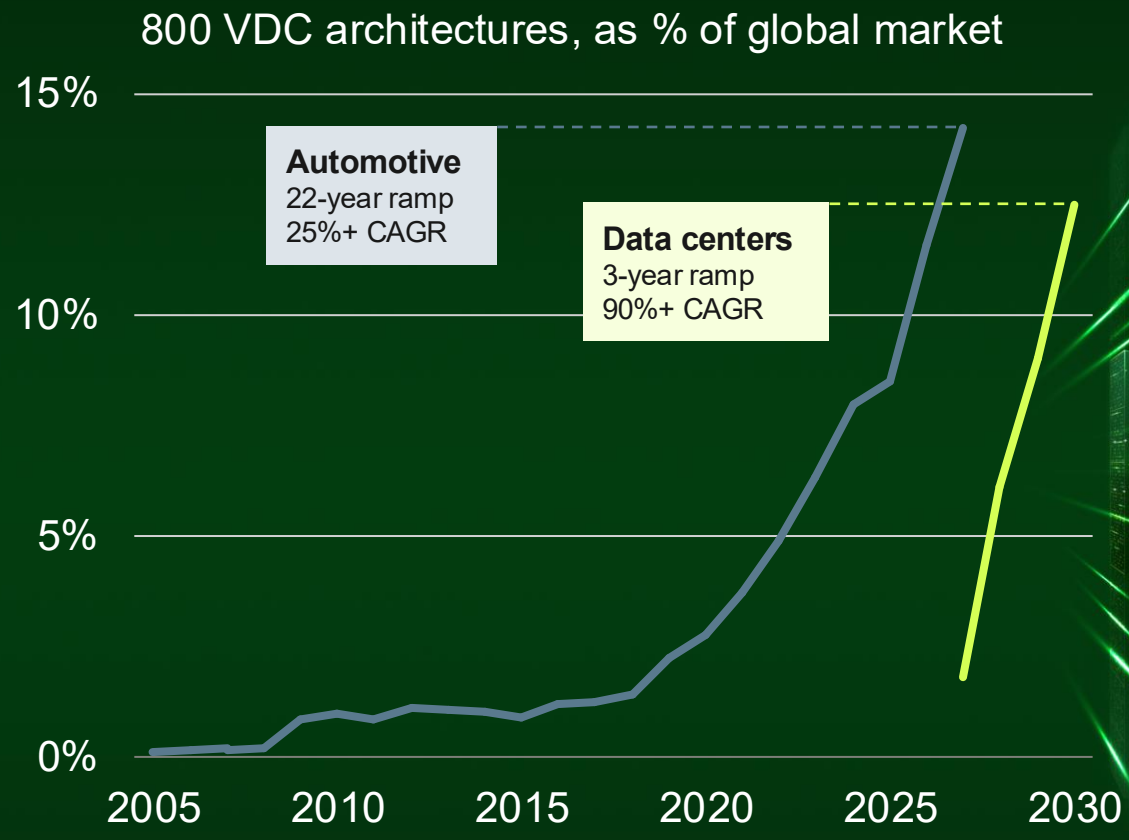
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CTO, Data Centers, Flex

OCP BRONZE™ Member

WHY 800 VDC AND WHY AUTOMOTIVE?

- 800 VDC is an inflection point in addressing the **power-density requirements** of AI data centers
- Automotive already innovated to solve problems across **power efficiency, safety controls, thermal performance, manufacturing processes**, and more
- **We can use this industry experience to accelerate 800 VDC development for data centers**



Source: S&P Mobility (May 2026), JLL (January 2026), Flex analysis (May 2026)

THE TOPOLOGIES AND TECHNOLOGIES ALIGN

800 VDC in automotive → 22 years to ramp, Voltages limited by torque

3 years to ramp, Voltage increases planned ← **800 VDC in data centers**

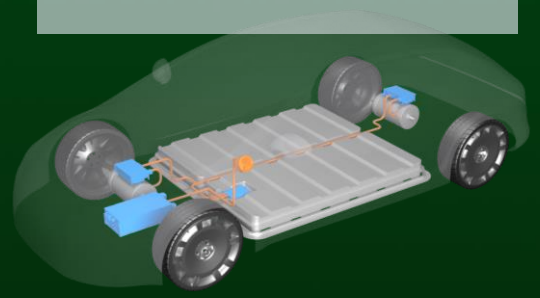
	Legacy (48 V mild hybrid)	**Modern** (800 V electric vehicle)	**Future** (800 VDC racks)	**Traditional** (>48 VAC racks)
POWER SOURCE	• Limited power via battery	• Limited power via battery	• Unlimited power via grid • Battery backup unit • Rectifier or SST	• Unlimited power via grid
POWER TRANSMISSION	• Cables and wiring harnesses	• EVSE cables • Interlock loop monitoring	• Switchgear • Busway and cables	• Switchgear • Busway and cables
POWER ELECTRONICS	• Power distribution unit • 48 V AC/DC inverter • 48 V-12 V DC/DC converter	• Power distribution unit • 800 V AC/DC inverter • 800 V-12 V DC/DC converter	• 800 V-48 V DC/DC converter • Step-down converters	• Power distribution unit • 48 V power shelf • Step-down converters

800 VDC architectures with cross-industry learning opportunities

Ramp data sources: S&P Mobility (May 2026), JLL (January 2026), Flex analysis (May 2026)

LEARNING OPPORTUNITIES: POWER SOURCE

Automotive	Data center
• Limited power via battery	• Unlimited power via grid • Battery backup unit • Rectifier or SST
• EVSE cables • Interlock loop monitoring	• Switchgear • Busway and cables
• Power distribution unit • 800 V AC/DC inverter • 800 V-12 V DC/DC converter	• 800 V-48 V DC/DC converter • Step-down converters





GRID INTERACTION

EVs are routinely charged from the grid with systems that are conceptually similar to utility and data center equipment

BATTERY MANAGEMENT AND SAFETY

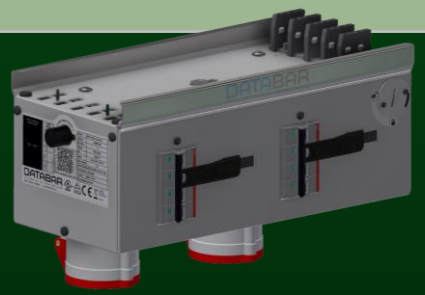
Lithium-ion batteries can be vulnerable to thermal runaway, and automotive has significant experience with this safety risk

A battery management system (BMS) and cooling systems are essential to support safe and reliable battery operation

LEARNING OPPORTUNITIES: POWER TRANSMISSION

Automotive	Data center
• Limited power via battery	• Unlimited power via the grid • Battery backup unit • Rectifier or SST
• **EVSE cables** • **Interlock loop monitoring**	• **Switchgear** • **Busway and cables**
• Power distribution unit • 800 V AC/DC inverter • 800 V-12 V DC/DC converter	• 800 V-48 V DC/DC converter • Step-down converters





CONNECTORS AND INTERLOCK LOOPS

Electric vehicle supply equipment (EVSE) incorporates multiple layers of protection

Interlock loops are designed to detect unintended access, open circuits, or non-standard operating parameters

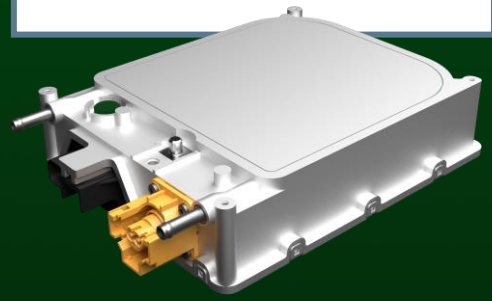
MAINTENANCE AND PERSONNEL

EV and data center maintenance personnel are both qualified and trained

Data center equipment often must remain energized to maintain system uptime

LEARNING OPPORTUNITIES: POWER ELECTRONICS

Automotive	Data center
• Limited power via battery	• Unlimited power via the grid • Battery backup unit • Rectifier or SST
• EVSE cables • Interlock loop monitoring	• Switchgear • Busway and cables
• **Power distribution unit** • **800 V AC/DC inverter** • **800 V-12 V DC/DC converter**	• **800 V-48 V DC/DC converter** • **Step-down converters**





INSULATION MONITORING

In an EV, an insulation monitoring device (IMD) continuously measures resistance between the 800 V bus and chassis

Unlike GFCIs or RCDs, an IMD is less susceptible to nuisance trips

SAFETY AND RELIABILITY

When a short-circuit fault occurs, EVs must transition to a safe state in <10 µs

EVs must ensure safety for failures below 100 FIT (10^9 operating hours)

CALL TO ACTION

- Read the full white paper, in collaboration with UL Solutions: https://flex.com/industries/data-center#resources
- Don’t reinvent the wheel, let’s leverage what those that came before us have innovated on
- Examine other opportunities for learnings, from automotive and across other industries
- Standards bodies should align data center requirements to existing requirements in other industries, where applicable
- Contribute to the OCP Power Distribution sub-project: https://www.opencompute.org/community/power-distribution

The potential benefits are **significant**

- > DEPLOYMENT TIMELINES
- > SAFETY AND RELIABILITY
- > UPTIME AND MONITORING
- > LONG-TERM SCALABILITY


Thank
You!
OCP
GLOBAL
SUMMIT
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